Exercise J-M

1. The number of points on the curve $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ at which the normal lines are parallel to $x + 90y + 2 = 0$ is : **[JEE(Main)Jan.2023]**

वक्र $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ पर उन बिंदुओं, जिन पर अभिलंब, रेखा $x + 90y + 2 = 0$ के समांतर है, की l [k; kgS **[JEE(Main)Jan.2023]**

(A) 3 (B) 0 (C) 4 (D) 2

Ans. C

Sol. Normal of the line is parallel to line $x + 90y + 2 = 0$

$$m_N = -\frac{1}{90}$$

$$-\left(\frac{dx}{dy}\right)_{(x_1y_1)} = -\frac{1}{90} \Rightarrow \left(\frac{dy}{dx}\right)_{(x_1y_1)} = 90$$

Now,

$$\frac{dy}{dx} = 270x^4 - 540x^3 - 210x^2 + 360x + 210 = 90$$

$$\Rightarrow x = 1, 2, \frac{-2}{3}, \frac{-1}{3}$$

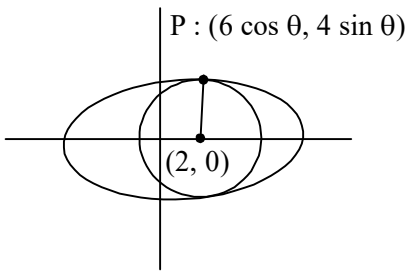
(D) normals

2. Let C be the largest circle centred at (2, 0) and inscribed in the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If $(1, \alpha)$ lies on C, then $10\alpha^2$ is equal to _____. **[JEE(Main)Jan.2023]**

माना C सबसे बड़ा वृत्त है ftldk dsUnz (2, 0) gS rFkk tks nh?kZo`Ùk $\frac{x^2}{36} + \frac{y^2}{16} = 1$ ds vUrxZr gSA ;fn fcUnq $(1, \alpha)$ o`Ùk C ij gS, rks $10\alpha^2$ cjkcj gS _____ **[JEE(Main)Jan.2023]**

Ans. 118

Sol.



equation of ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$

$$\frac{x}{18}+\frac{yy'}{8}=0$$

$$y'=\frac{-x}{18}.\frac{8}{y}$$

$$Y_N=\frac{9y}{4x}=9\cdot\frac{4\sin\theta}{4\cdot 6\cos\theta}=\frac{3}{2}\frac{\sin\theta}{\cos\theta}$$

$$Y_N(\text{ Slope })=\frac{4\sin\theta}{6\cos\theta-2}=\frac{2\sin\theta}{3\cos\theta-1}$$

$$\frac{3}{2}\frac{\sin\theta}{\cos\theta}=\frac{2\sin\theta}{3\cos\theta-1}$$

$$\cos\theta=\frac{3}{5}$$

$$P:\left(\frac{18}{5},\frac{16}{5}\right)$$

$$S=\sqrt{\left(\frac{8}{5}\right)^2+\left(\frac{16}{5}\right)^2}$$

Now $(x-2)^2+y^2=r^2$

$$(1,\alpha)\to 1+\alpha^2=\frac{8^2}{5^2}+\frac{16^2}{5^2}=12.8$$

$\alpha^2 = 11.8$
$10\alpha^2 = 118$

3. If the equation of the normal to the curve $y=\frac{x-a}{(x+b)(x-2)}$ at the point (1,–3) is $x-4y=13$, then the value of $a+b$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि वक्र $y=\frac{x-a}{(x+b)(x-2)}$ के बिंदु (1,-3) पर अभिलंब का समीकरण x - $4y$ = 13 है, तो $x=\frac{5}{2}$ पर $a+b$ का मान बराबर है।

[JEE(Main)Jan.2023]

Ans. 4

Sol. $y=\frac{x-a}{(x+b)(x-2)}$

At point (1, –3),

$$-3=\frac{1-9}{(1+b)(1-2)}$$

$\Rightarrow 1 - a = 3(1 + b)$ ...(1)

Now, $y=\frac{x-a}{(x+b)(x-2)}$

$$\Rightarrow \frac{dy}{dx} = \frac{(x+b)(x-2)\times(1)-(x-a)(2x+b-2)}{(x+b)^2(x-2)^2}$$

At (1, –3) slope of normal is $\frac{1}{4}$ hence $\frac{dy}{dx} = -4$,

So, $-4 = \frac{(1+b)(-1)-(1-a)b}{(1+b)^2(-1)^2}$

Using equation (1)

$$\Rightarrow -4 = \frac{(1+b)(-1)-3(b+1)b}{(1+b)^2}$$

$$\Rightarrow -4 = \left|\frac{(-1)-3b}{(1+b)}\right| (b \neq -1)$$

$\Rightarrow b = -3$

So, $a = 7$

Hence, $a + b = 7 - 3 = 4$

4. Let $f: (0, 1) \to R$ be a function defined by $f(x) = \frac{1}{1-e^{-x}}$, and $g(x) = (f(-x) - f(x))$. Consider two statements

(I) g is an increasing function in (0, 1)
(II) g is one-one in (0, 1)

Then, : **[JEE(Main)Jan.2023]**

(A) Both (I) and (II) are true (B) Only (I) is true

(C) Neither (I) nor (II) is true (D) Only (II) is true

माना $f: (0, 1) \to R$, $f(x) = \frac{1}{1-e^{-x}}$ द्वारा परिभाषित है तथा $g(x) = (f(-x) - f(x))$ है। दो कथनों का विचार कीजिए।

(I) (0, 1) में g एक वर्धमान फलन है
(II) (0, 1) में g एकैकी है

r ks: **[JEE(Main)Jan.2023]**

(A) (I) तथा (II) दोनों सत्य है (B) केवल (I) सत्य है

(C) न तो (I) न ही (II) सत्य है (D) केवल (II) सत्य है

Ans. A

Sol. $g(x) = f(-x) - f(x) = \frac{1}{1-e^{x}} - \frac{1}{1-e^{-x}}$

$$\Rightarrow g(x) = \frac{1+e^{x}}{1-e^{x}}$$

$$g'(x) = \frac{\left(1-e^{x}\right)\left(e^{x}\right)-\left(1+e^{x}\right)\left(-e^{x}\right)}{\left(1-e^{x}\right)^2}$$

$$g'(x)=\frac{2e^x}{\left(1-e^x\right)^2}>0$$

g (x) is increasing as well as one-one.

5. Let $f:\ R\to R$ be a function such that $f(x)=\frac{x^2+2x+1}{x^2+1}$. Then **[JEE(Main)Jan.2023]**

(A) $f(x)$ is one-one in $[1,\infty)$ but not in $(-\infty,\infty)$

(B) $f(x)$ is one-one in $(-\infty,\infty)$

(C) $f(x)$ is many-one in $(1,\infty)$

(D) $f(x)$ is many-one in $(-\infty,-1)$

माना $f:\ R\to R$ एक फलन $f(x)=\frac{x^2+2x+1}{x^2+1}$ है। तब **[JEE(Main)Jan.2023]**

(A) $[1,\infty)$ में $f(x)$ एकैकी है परन्तु $(-\infty,\infty)$ में एकैकी नहीं है।

(B) $(-\infty,\infty)$ में $f(x)$ एकैकी है।

(C) $(1,\infty)$ में $f(x)$ बहु–एक है।

(D) $(-\infty,-1)$ में $f(x)$ बहु–एक है।

Ans. A

Sol. $\because f(x)=\frac{(x+1)^2}{(x+1)}$

$$\because f'(x)=\frac{\left(x^2+1\right).2(x+1)-(x+1)^2(2x)}{\left(x^2+1\right)^2}=-\frac{2(x+1)(x-1)}{\left(x^2+1\right)^2}$$

clearly f(x) is one – one is $(-\infty, -1)$ and also in $(1, \infty)$ but f(x) is not one – one is $(-\infty, \infty)$

6. Let f and g be twice differentiable functions on R such that

$$f''(x)=g''(x)+6x,\quad f'(1)=4g'(1)-3=9,\quad f(2)=3g(2)=12$$

Then which of the following is NOT true? **[JEE(Main)Jan.2023]**

(A) $g(-2)-f(-2)=20$

(B) If $-1<x<2$, then $|f(x)-g(x)|<8$

(C) There exists $x_0\in(1,3/2)$ such that $f(x_0)=g(x_0)$

(D) $|f'(x)-g'(x)|<6\Rightarrow -1<x<1$

माना फलन f तथा g, R पर दो बार अवकलनीय हैं तथा

$$f''(x)=g''(x)+6x,\quad f'(1)=4g'(1)-3=9,\quad f(2)=3g(2)=12$$

हैं। तो निम्न में से कौन सा सत्य नहीं है ? **[JEE(Main)Jan.2023]**

(A) $g(-2) - f(-2) = 20$ है

(B) यदि $-1 < x < 2$ है, तो $|f(x) - g(x)| < 8$ है

(C) $x_0 \in (1, 3/2)$ का अस्तित्व है, जिसके लिए $f(x_0) = g(x_0)$ है

(D) $|f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$

Ans. B

Sol. $f''(x) = g''(x) + 6x$

$\Rightarrow \int f''(x)dx = \int g''(x)dx + \int 6xdx + C$

$\Rightarrow f'(x) = g'(x) + 3x^2 + C$ .....(1)

$x = 1 \Rightarrow f'(1) = g'(1) + 3 + C$

$\Rightarrow 9 = 3 + 3 + C \Rightarrow C = 3$

from (1) $\int f'(x)dx = \int g'(x)dx + \int 3x^2dx + \int 3dx + k$

$\Rightarrow f(x) = g(x) + x^3 + 3x + K$

$x = 2 \Rightarrow f(2) = g(2) + 8 + 6 + K$

$\Rightarrow 12 = 4 + 8 + 6 + K$

$\Rightarrow K = -6$

$\Rightarrow f(x) = g(x) + x^3 + 3x - 6$

Let $h(x) = f(x) - g(x) = x^3 + 3x - 6$

$\Rightarrow h'(x) = f'(x) - g'(x) = 3x^2 + 3 > 0 \ \forall \ x$

$\Rightarrow h'(x)$ is S.I. $\Rightarrow h(x) = 0$ has no solution

$\Rightarrow h(x) \in (h(-1), h(2)) \Rightarrow h(x) \in (-8, 8)$

$|f'(x) - g'(x)| < 6 \Rightarrow |3x^2 + 3| < 6 \Rightarrow 3x^2 + 3 < 6$

$\Rightarrow x^2 - 1 < 0 \Rightarrow x \in (-1, 1)$

$h(-2) = f(-2) - g(-2) = -20$

7. Let $x = 2$ be local minima of the function $f(x) = 2x^4 - 18x^2 + 8x + 12$, $x \in (-4, 4)$. If M is local maximum value of the function f in $(-4, 4)$, then M = **[JEE(Main)Jan.2023]**

माना फलन $f(x) = 2x^4 - 18x^2 + 8x + 12$, $x \in (-4, 4)$ का एक स्थानीय निम्नतम $x = 2$ पर है। यदि $(-4, 4)$ में फलन f का स्थानीय महत्तम M है, तो M = **[JEE(Main)Jan.2023]**

(A) $12\sqrt{6} - \frac{33}{2}$ (B) $18\sqrt{6} - \frac{31}{2}$ (C) $18\sqrt{6} - \frac{33}{2}$ (D) $12\sqrt{6} - \frac{31}{2}$

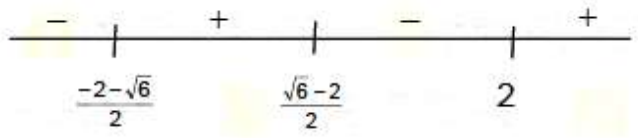
Ans. A

Sol. $f(x) = 2x^4 - 18x^2 + 8x + 12$

$f'(x) = 8x^3 - 36x + 8$

$= 4(x - 2)(2x^2 + 4x - 1)$

Number line of $f'(x)$ is given by



so maxima point is $x = \frac{\sqrt{6}-2}{2}$

and maximum value of $f(x) = 12\sqrt{6} - \frac{33}{2}$

8. Let the function $f(x) = 2x^3 + (2p - 7)x^2 + 3(2p - 9)x - 6$ have a maxima for some value of $x < 0$ and a minima for some value of $x > 0$. Then, the set of all values of p is : **[JEE(Main)Jan.2023]**

माना फलन $f(x) = 2x^3 + (2p - 7)x^2 + 3(2p - 9)x - 6$ का एक उच्चिष्ठ किसी $x < 0$ पर है तथा एक निम्निष्ठ किसी $x > 0$ पर है। तो p के सभी मानों का समुच्चय है **[JEE(Main)Jan.2023]**

(A) $\left(\frac{9}{2}, \infty\right)$ (B) $\left(-\infty, \frac{9}{2}\right)$ (C) $\left(-\frac{9}{2}, \frac{9}{2}\right)$ (D) $\left(0, \frac{9}{2}\right)$

Ans. B

Sol. $f'(x) = 6x^2 + 2(2p-7)x + 3(2p-9)$

$f'(0) < 0$

$3(2p-9) < 0$

$2p - 9 < 0$

$p < \frac{9}{2} \Rightarrow p \in \left(-\infty, \frac{9}{2}\right)$

9. If the functions $f(x) = \frac{x^3}{3} + 2bx + \frac{ax^2}{2}$ and $g(x) = \frac{x^3}{3} + ax + bx^2$, $a \neq 2b$ have a common extreme point, then $a + 2b + 7$ is equal to : **[JEE(Main)Jan.2023]**

यदि फलनों $f(x) = \frac{x^3}{3} + 2bx + \frac{ax^2}{2}$ तथा $g(x) = \frac{x^3}{3} + ax + bx^2$, $a \neq 2b$ का एक उभयनिष्ठ चरम बिंदु है, तब $a + 2b + 7$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 4 (B) $\frac{3}{2}$ (C) 6 (D) 3

Ans. C

Sol. $f'(x) = x^2 + 2b + ax$

$g'(x) = x^2 + a + 2bx$

$(2b - a) - x(2b - a) = 0$

$\therefore x = 1$ is the common root

Put $x = 1$ in $f'(x) = 0$ or $g'(x) = 0$

$1 + 2b + a = 0$

$7 + 2b + a = 6$

10. A wire of length 20 m is to be cut into two pieces. A piece of length l_1 is bent to make a square of area A_1 and the other piece of length l_2 is made into a circle of area A_2. If $2A_1 + 3A_2$ is minimum then $(\pi l_1) : l_2$ is equal to : **[JEE(Main)Jan.2023]**

20 मीटर लंबी एक तार को दो भागों में काटा गया है। l_1 लंबाई के एक भाग को मोड़ कर A_1 क्षेत्रफल का एक वर्ग बनाया गया है तथा l_2 लंबाई के दूसरे भाग से A_2 क्षेत्रफल का एक वृत्त बनाया गया है। यदि $2A_1 + 3A_2$ न्यूनतम है, तो $(\pi l_1) : l_2$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 4 : 1 (B) 1 : 6 (C) 6 : 1 (D) 3 : 1

Ans. C

Sol. $\ell_1 + \ell_2 = 20 \Rightarrow \dfrac{d\ell_2}{d\ell_1} = -1$

$A_1 = \left(\dfrac{\ell_1}{4}\right)^2$ and $A_2 = \pi\left(\dfrac{\ell_2}{2\pi}\right)^2$

Let $S = 2A_1 + 3A_2 = \dfrac{\ell_1^2}{8} + \dfrac{\ell_2^2}{4\pi}$

$\dfrac{ds}{d\ell} = 0 \Rightarrow \dfrac{2\ell_1}{8} + \dfrac{6\ell_2}{4\pi} \cdot \dfrac{d\ell_2}{d\ell_1} = 0$

$\Rightarrow \dfrac{\ell_1}{4} = \dfrac{6\ell_2}{4\pi} \Rightarrow \dfrac{\pi\ell_1}{\ell_2} = 6$

11. Let $f(x) = 2x + \tan^{-1} x$ and $g(x) = \log_e\left(\sqrt{1+x^2} + x\right)$, $x \in [0, 3]$. Then : **[JEE(Main)Jan.2023]**

(A) $\min f'(x) = 1 + \max g'(x)$

(B) there exist $0 < x_1 < x_2 < 3$ such that $f(x) < g(x)$, $\forall\ x \in (x_1, x_2)$

(C) there exists $\hat{x} \in [0,3]$ such that $f'(\hat{x}) < g'(\hat{x})$

(D) $\max f(x) > \max g(x)$

माना $f(x) = 2x + \tan^{-1} x$ एवं $g(x) = \log_e\left(\sqrt{1+x^2} + x\right)$, $x \in [0, 3]$ है। तब **[JEE(Main)Jan.2023]**

(A) $\min f'(x) = 1 + \max g'(x)$

(B) $0 < x_1 < x_2 < 3$ का अस्तित्व है, जिनके लिए $f(x) < g(x)$, $\forall\ x \in (x_1, x_2)$ है।

(C) $\hat{x} \in [0,3]$ का अस्तित्व है, जिसके लिए $f'(\hat{x}) < g'(\hat{x})$ है।

(D) $\max f(x) > \max g(x)$

Ans. D

Sol. $f(x)=2x+\tan^{-1}x$ and $g(x)=\ln\left(\sqrt{1+x^2}+x\right)$

and $x \in [0, 3]$

$$g'(x)=\frac{1}{\sqrt{1+x^2}}$$

Now, $0 \le x \le 3$

$0 \le x^2 \le 9$

$1 \le 1 + x^2 \le 10$

So, $2+\frac{1}{10} \le f'(x) \le 3$

$\frac{21}{10} \le f'(x) \le 3$ and $\frac{1}{\sqrt{10}} \le g'(x) \le 1$

option (4) is icorrect

From above, $g'(x) < f'(x)\ \forall\ x \in [0, 3]$

Option (1) is incorrect

f'(x) & g'(x) both positive so f(x) & g(x) both are increasing

So, max (f(x) at x = 3 is $6 + \tan^{-1} 3$)

Max (g(x) at x = 3 is $\ln\left(3+\sqrt{10}\right)$

And $6 + \tan^{-1} 3 > \ln\left(3+\sqrt{10}\right)$

Option (4) is correct.

12. The sum of the absolute maximum and minimum values of the function $f(x) = |x^2 - 5x + 6| - 3x + 2$ in the interval [-1,3] is equal to : **[JEE(Main)Jan.2023]**

अंतराल [-1,3] में फलन $f(x) = |x^2 - 5x + 6| - 3x + 2$ के निरपेक्ष उच्चतम और निरपेक्ष निम्नतम मानों का योग बराबर है:

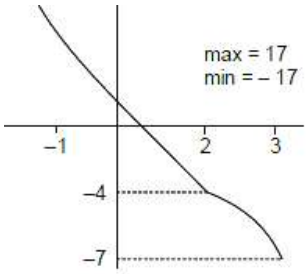
[JEE(Main)Jan.2023]

(A) 12 (B) 24 (C) 10 (D) 13

Ans. C

Sol. $f(x) = |x^2 - 5x + 6| - 3x + 2$

$$f(x)=\begin{cases} x^2-8x+8 & ; \quad x[-1,2] \\ -x^2+2x-4 & ; \quad x \in [2,3] \end{cases}$$



$$f'(x)=\begin{cases} 2x-8; & x\in[-1,2] \\ -2x+2; & x\in[2,3] \end{cases}$$

$f(2) = 4 - 16 + 8 = -4 = 4$

$f(-1) = 17$ (max)

$f(3) = -9 + 6 - 4 = -7$ (min)

Abs. Sum = 10

13. Let a tangent to the curve $9x^2 + 16y^2 = 144$ intersects the coordinate axes at the points A and B. Then, the minimum length of the line segment AB is _____. **[JEE(Main)Jan.2023]**

माना वक्र $9x^2 + 16y^2 = 144$ की एक स्पर्श रेखा निर्देशांक अक्षों को बिंदुओं A तथा B पर मिलती है। तो रेखाखंड AB की न्यूनतम लंबाई gS_____ **[JEE(Main)Jan.2023]**

Ans. 7

Sol. $\frac{x^2}{16}+\frac{y^2}{9}=1$

Equation of tangent $\frac{x}{4}\cos\theta+\frac{y}{3}\sin\theta=1$

A (4sec θ, 0) B(0, 3(cosec θ)

$$AB=\sqrt{16\sec^2\theta+9\operatorname{cosec}^2\theta}=\sqrt{25+\left(4\tan\theta-3\cot\theta\right)^2+24}\geq\sqrt{49}\geq 7$$

$AB_{min} = 7$

14. Let the quadratic curve passing through the point (–1, 0) and touching the line y = x at (1, 1) be y = f(x). Then the x-intercept of the normal to the curve at the point $(\alpha, \alpha + 1)$ in the first quadrant is _____. **[JEE(Main)April,2023]**

माना बिंदु (–1, 0) से होकर जाने वाला तथा रेखा y = x को (1, 1) पर स्पर्श करने वाला द्विघातीय वक्र y = f(x) है। तो प्रथम चतुर्थांश esfcaq$(\alpha, \alpha + 1)$ पर वक्र के अभिलंब का x-अंतःखंड है : **[JEE(Main)April,2023]**

Ans. 11

Sol. $f(x) = (x+1)(ax+b)$

$1 = 2a + 2b$ ....(1)

$f'(x) = (ax+b) + a(x+1)$

$1 = (3a + b)$ ....(2)

$\Rightarrow b = 1/4, a = 1/4$

$$f(x)=\frac{(x+1)^2}{4}$$

$$f'(x)=\frac{x}{2}+\frac{1}{2}\quad \alpha+1=\frac{(\alpha+1)^2}{4},\alpha>-1$$

$\alpha + 1 = 4$

$\alpha = 3$

normal at (3, 4)

$y-4=-\frac{1}{2}(x-3)$

$y=0 \qquad x=8+3$

15. Let $g(x)=f(x)+f(1-x)$ and $f''(x)>0, x\in(0,1)$. If g is decreasing in the interval $(0, \alpha)$ and increasing in the interval $(\alpha, 1)$, then $\tan^{-1}(2\alpha)+\tan^{-1}\left(\frac{1}{\alpha}\right)+\tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$ is equal to : **[JEE(Main)April,2023]**

माना $g(x)=f(x)+f(1-x)$ तथा $f''(x)>0, x\in(0,1)$ है। यदि g अंतराल $(0, \alpha)$ में ह्यसमान है तथा अंतराल $(\alpha, 1)$ में वर्धमान है, r ks$\tan^{-1}(2\alpha)+\tan^{-1}\left(\frac{1}{\alpha}\right)+\tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$ बराबर है : **[JEE(Main)April,2023]**

(A) π (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{3\pi}{2}$

Ans. A

Sol. $g'(x)=f'(x)-f'(1-x)$

If $x>1-x$

$\Rightarrow f'(x)>f'(1-x) \because f'(x)$ is strictly increasing

$\Rightarrow g'(x)>0 \Rightarrow$ so g(x) is increasing when $x>1-x \Rightarrow x>\frac{1}{2}$

$\Rightarrow \alpha=\frac{1}{2}$

similarly when $x<1-x \Rightarrow g(x)$ is decreeing

Now

$\tan^{-1}(2\alpha)+\tan^{-1}\left(\frac{1}{\alpha}\right)+\tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$

$=\tan^{-1}(1)+\tan^{-1}(2)+\tan^{-1}(3)=\pi(\because 1+2+3=1.2.3)$

16. Let $f:[2,4]\to\mathbb{R}$ be a differentiable function such that

$(x\log_e x)f'(x)+(\log_e x)f(x)+f(x)\geq 1, x\in[2,4]$ with $f(2)=\frac{1}{2}$ and $f(4)=\frac{1}{4}$.

Consider the following two statements :

(A) : $f(x)\leq 1$, for all $x\in[2, 4]$

(B) : $f(x)\geq\frac{1}{8}$, for all $x\in[2, 4]$

Then : **[JEE(Main)April,2023]**

(A) Only statement (A) is true (B) Both the statements (A) and (B) are true

(C) Neither statement (A) nor statement (B) is true (D) Only statement (B) is true

माना $f:[2,4]\to\mathbb{R}$ एक अवकलनीय फलन है, जिसके लिए

$(x\log_e x)f'(x)+(\log_e x)f(x)+f(x)\ge 1, x\in[2,4]$, $f(2)=\frac{1}{2}$ तथा $f(4)=\frac{1}{4}$ है।

fuEu esa ls dkSu&ls Fkus dFku lR; gSa ?

; । द्वारा सभी $x\in[2,4]$ के लिए $f(x)\le 1$ है।

(B) : सभी $x\in[2,4]$ के लिए $f(x)\ge\frac{1}{8}$ है।

rks **[JEE(Main)April,2023]**

(A) केवल कथन (A) सत्य है (B) (A) तथा (B) दोनों कथन सत्य हैं

(C) न तो कथन (A) न ही कथन (B) सत्य है (D) केवल कथन (B) सत्य है

Ans. B

Sol. $\frac{d}{dx}\left(x\,\ell n(x).f(x)-x\right)\ge 0,\forall x\in[2,4]$

let $g(x)=x\,\ell n(x)f(x)-x,\quad \forall x\in[2,4]$

$\Rightarrow g'(x)\ge 0,\forall x\in[2,4]$

$\Rightarrow g'(x)$ is increasing funtion in [2,4]

$\Rightarrow g(2)\le g(x)\le g(4)$

$\Rightarrow 2\cdot\ell n\cdot\frac{1}{2}-2\le x\,\ell n(x)f(x)-x\le 4\cdot\ell n(4)\cdot\frac{1}{4}-4$

$\Rightarrow \ln 2-2+x\le x.\ell nx.f(x)\le 2(\ell n2-2)+x$

$\Rightarrow \frac{\ell n2-2}{x\ell nx}+\frac{1}{\ell nx}\le f(x)\le\frac{2(\ln 2-2)}{x\ell nx}+\frac{1}{\ell nx}$

Since numerator is constant

$\Rightarrow \frac{\ell n2-2}{4\ell n4}+\frac{1}{\ell\ln 4}\le f(x)\le\frac{2(\ell n2-2)}{2\ell n2}+\frac{1}{\ell n2}$

$\Rightarrow \frac{1}{8}+\frac{1}{4\ln 2}\le f(x)\le 1-\frac{1}{\ell n2}$

$2\Rightarrow \frac{1}{8}<f(x)<1,\forall x\in[2,4]$

17. The number of points, where the curve $y=x^5-20x^3+50x+2$ crosses the x-axis, is _____.

[JEE(Main)April,2023]

उन बिंदुओं, जहाँ वक्र $y=x^5-20x^3+50x+2$, x-अक्ष को काटता है, की संख्या है _____

[JEE(Main)April,2023]

Ans. 5

Sol. $y = x^5 - 20x^3 + 50x + 2$

$$\frac{dy}{dx} = 5x^4 - 60x^2 + 50 = 5\left(x^4 - 12x^2 + 10\right)$$

$$\frac{dy}{dx} = 0 \Rightarrow x^4 - 12x^2 + 10 = 0$$

$$\Rightarrow x^2 = \frac{12 \pm \sqrt{144 - 40}}{2}$$

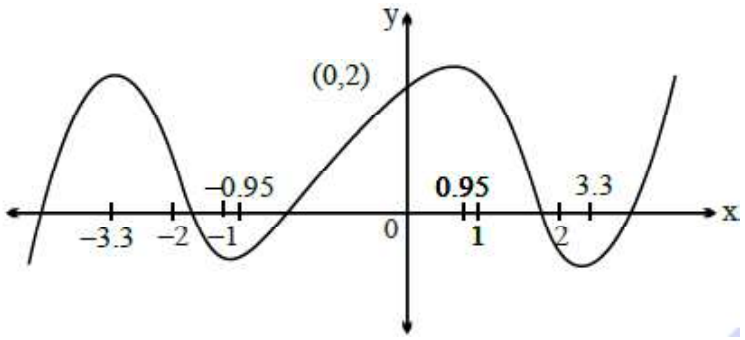
$$\Rightarrow x^2 = 6 \pm \sqrt{26} \Rightarrow x^2 \approx 6 \pm 5.1$$

$$\Rightarrow x^2 \approx 11.1, 0.9$$

$$\Rightarrow x \approx \pm 3.3, \pm 0.95$$

$f(0) = 2, f(1) = +\text{ ve}, f(2) = -\text{ ve}$

$f(-1) = -\text{ve}, f(-2) = +\text{ve}$



18. If a_α is the greatest term in the sequence $a_n = \frac{n^3}{n^4 + 147}, n = 1, 2, 3, \ldots$ then α is equal to _____.

[JEE(Main)April,2023]

यदि अनुक्रम $a_n = \frac{n^3}{n^4 + 147}, n = 1, 2, 3, \ldots$ का अधिकतम पद a_α है, तो α बराबर है _____

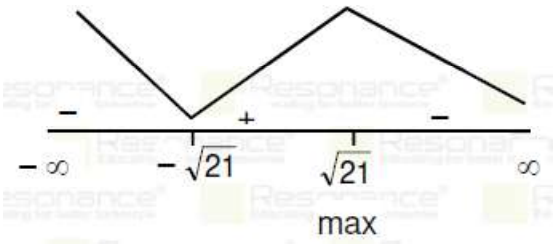
[JEE(Main)April,2023]

Ans. 5

Sol. $$f'(n) = \frac{3n^2\left(n^4 + 147\right) - 4n^3\left(n^3\right)}{\left(n^4 + 147\right)^2}$$

$$f'(n) = \frac{n^2\left[3n^4 + 441 - 4n^4\right]}{\left(n^4 + 147\right)^2}$$

$$f'(n) = \frac{-n^2\left(n^4 - 441\right)}{\left(n^4 + 147\right)^2} = \frac{-n^2\left(n^2 + 21\right)\left(n^2 - 21\right)}{\left(n^4 + 147\right)^2} = \frac{-n^2\left(n^2 + 21\right)}{\left(n^4 + 147\right)^2}(n + \sqrt{21})(n - \sqrt{21})$$



When $n = 4 \quad a_4 = \frac{4^3}{4^4 + 147} = \frac{64}{256 + 147} = \frac{64}{403}$

When $n = 5 \quad a_5 = \frac{125}{625 + 147} = \frac{125}{772}$

a_5 is $\max = \frac{125}{772}$ hence $n = 5$

19. The number of points, where the curve $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$ cuts x-axis, is equal to _____.

उन बिंदुओं, जहाँ वक्र $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$, x-अक्ष को काटता है, की संख्या है _____

Ans. B

Sol. $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$

$\lim_{x \to -\infty} f(x) = 1, \quad f(0) = -3, \quad \lim_{x \to \infty} f(x) \to \infty$

One root of f(x) = 0 is negative and other root of f(x) = 0 is positive

Curve y = f(x) cuts x-axis at two points

20. A square piece of tin of side 30 cm is to be made into a box without top by cutting a square from each corner and folding up the flaps to form a box. If the volume of the box is maximum, then its surface area (in cm^2) is equal to :

[JEE(Main)April,2023]

30 cm भुजा के टिन के एक एक वर्गाकार टुकड़े के प्रत्येक कोने पर एक वर्ग काटकर तथा इस प्रकार बनें टिन के फलकों को मोड़ कर <Dl u j fgr , d l aoo cukukgSA; fn l aoo d kvk, r u mPpre gSr ksbl d ki "Bh {ks=kQy ¼oxZ cm में) बराबर है :

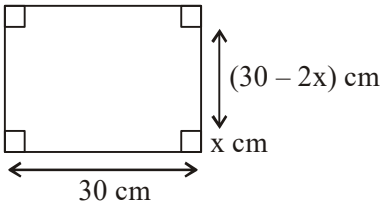
[JEE(Main)April,2023]

(A) 675 (B) 1025 (C) 900 (D) 800

Ans. D

Sol. Let the side of the square to be cut off be x cm.

Then, the length and breadth of the box will be (30 – 2x) cm each and the height of the box is x cm therefore,



The volume V(x) of the box is given by

$V(x) = x(30 - 2x)^2$

$\frac{dv}{dx} = (30 - 2x)^2 + 2x(30 - 2x)(-2)$

$0 = (30 - 2x)^2 - 4x(30 - 2x)$

$0 = (30 - 2x)[(30 - 2x) - 4x]$

$0 = (30 - 2x)(30 - 6x)$

$x = 15, 5$

$x \neq 15$ (Not possible)

$\{V = 0\}$

Surface area without top of the box $= \ell b + 2(bh + h\ell)$

$= (30 - 2x)(30 - 2x) + 2[(30 - 2x)x + (30 - 2x)x]$

$= [(30 - 2x)^2 + 4\{(30 - 2x)x\}$

$= [(30 - 10)^2 + 4(5)(30 - 10)]$

$= 400 + 400$

$= 800\ cm^2$

21. If the local maximum value of the function $f(x) = \left(\frac{\sqrt{3}e}{2\sin x}\right)^{\sin^2 x}$, $x \in \left(0, \frac{\pi}{2}\right)$, is $\frac{k}{e}$, then $\left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8$ is equal to : **[JEE(Main)April,2023]**

यदि फलन $f(x) = \left(\frac{\sqrt{3}e}{2\sin x}\right)^{\sin^2 x}$, $x \in \left(0, \frac{\pi}{2}\right)$, का स्थानीय उच्चतम मान $\frac{k}{e}$ है, तो $\left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8$ बराबर है : **[JEE(Main)April,2023]**

(A) $e^5 + e^6 + e^{11}$ (B) $e^3 + e^6 + e^{10}$ (C) $e^3 + e^5 + e^{11}$ (D) $e^3 + e^6 + e^{11}$

Ans. A

Sol. Let $y = \left(\frac{\sqrt{3}e}{2\sin x}\right)^{\sin^2 x}$

$$\ln y = \sin^2 x \cdot \ln\left(\frac{\sqrt{3}e}{2\sin x}\right)$$

$$\frac{1}{y}y' = \ln\left(\frac{\sqrt{3}e}{2\sin x}\right)2\sin x\cos x + \sin^2 x \frac{2\sin x}{\sqrt{3}e}\frac{\sqrt{3}e}{2}(-\operatorname{cosec} x \cot x)$$

$$\Rightarrow \sin x \cos\left[2\ln\left(\frac{\sqrt{3}e}{2\sin x}\right) - 1\right] = 0$$

$$\Rightarrow \ln\left(\frac{3e}{4\sin^2 x}\right) = 1 \Rightarrow \frac{3e}{4\sin^2 x} = e \Rightarrow \sin^2 x = \frac{3}{4}$$

$$\Rightarrow \sin x = \frac{\sqrt{3}}{2}\left(\text{as } x \in \left(0, \frac{\pi}{2}\right)\right)$$

$$\Rightarrow \text{local max value} = \left(\frac{\sqrt{3e}}{\sqrt{3}}\right)^{3/4} = e^{3/8} = \frac{k}{e}$$

$$\Rightarrow k^8 = e^{11}$$

$$\Rightarrow \left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8 = e^3 + e^6 + e^{11}$$

22. $\max_{0\le x\le \pi}\left\{x - 2\sin x\cos x + \frac{1}{3}\sin 3x\right\} =$ **[JEE(Main)April,2023]**

(A) π (B) $\frac{5\pi + 2 + 3\sqrt{3}}{6}$ (C) 0 (D) $\frac{\pi + 2 - 3\sqrt{3}}{6}$

Ans. B

Sol. $f(x) = x - \sin 2x + \frac{1}{3}\sin 3x$

$f'(x) = 1 - 2\cos 2x + \cos 3x = 0$

$x = \frac{5\pi}{6}, \frac{\pi}{6}$

$\therefore f''(x) = 4\sin 2x - 3\sin 3x$

$f''\left(\frac{5\pi}{6}\right) < 0$

$\Rightarrow \left(\frac{5\pi}{6}\right)$ is point of maxima

$\Rightarrow f\left(\frac{5\pi}{6}\right) = \frac{5\pi}{6} + \frac{\sqrt{3}}{2} + \frac{1}{3}$

23. Let $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ and $f''(x) > 0$ for all $x \in (0, 3)$. If g is decreasing in $(0, \alpha)$ and increasing in $(\alpha, 3)$, then 8α is : **[JEE(Main)Jan.2024]**

माना सभी $x \in (0, 3)$ के लिए $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ तथा $f''(x) > 0$ हैं। यदि $(0, \alpha)$ में g ह्रासमान है तथा $(\alpha, 3)$ में वध

kZku gS,Sतो 8α बराबर है : **[JEE(Main)Jan.2024]**

(A) 20 (B) 0 (C) 24 (D) 18

Ans. D

Sol. Since $f''(x) > 0 \Rightarrow f'(x)$
is increasing in $(0, 3)$

Now $g'(x) = f'\left(\frac{x}{3}\right) - f'(3-x)$

For g(x)

to be increasing $g'(x) = f'\left(\frac{x}{3}\right) - f'(3-x) > 0$

$\Rightarrow f'\left(\frac{x}{3}\right) > f'(3-x)$

$\Rightarrow \frac{x}{3} > 3-x$

$\Rightarrow x > 9-3x \Rightarrow x > \frac{9}{4}$

For g(x) be be decreasing

$g'(x) < 0 \quad \Rightarrow x < \frac{9}{4} \qquad \Rightarrow \alpha = \frac{9}{4}$

24. Consider the function $f : \left[\frac{1}{2}, 1\right] \to \mathbb{R}$ defined by $f(x) = 4\sqrt{2}x^3 - 3\sqrt{2}x - 1$. Consider the statements :

(I) The curve y = f(x) intersects the x-axis exactly at one point

(II) The curve y = f(x) intersects the x-axis at $x = \cos\frac{\pi}{12}$.

Then : **[JEE(Main)Jan.2024]**

(A) Only (I) is correct (B) Both (I) and (II) are the correct

(C) Both (I) and (II) are incorrect (D) Only (II) is correct

$f(x) = 4\sqrt{2}x^3 - 3\sqrt{2}x - 1$ द्वारा परिभाषित फलन $f : \left[\frac{1}{2}, 1\right] \to \mathbb{R}$ के लिए कथनों :

;प्द्धवक्र y = f(x), x-अक्ष को मात्र एक बिन्दु पर काटता है।

(II) वक्र y = f(x), x-अक्ष को $x = \cos\frac{\pi}{12}$ पर काटता है।

ed s **[JEE(Main)Jan.2024]**

(A) केवल (I) सही है (B) (I) तथा (II) दोनों सही हैं

(C) (I) तथा (II) दोनों गलत हैं (D) केवल (II) सही हैं

Ans. B

Sol. Let $x = \frac{\pi}{12}$

$\therefore 3x = \frac{\pi}{4}$

$\cos(3x) = \frac{1}{\sqrt{2}}$

$\Rightarrow 4\cos^3 x - 3\cos x = \frac{1}{\sqrt{2}}$

$\Rightarrow 4\sqrt{2}\cos^3 x - 3\sqrt{2}\cos x - 1 = 0$

$\therefore x = \left(\frac{\pi}{12}\right)$ is the solution of above equation statement II is correct

$f'(x) = 12\sqrt{2}x^2 - 3\sqrt{2}$

Which is positive for $x \in \left[\frac{1}{2}, 1\right]$

Hence, only one solution in $\left[\frac{1}{2}, 1\right]$

$\therefore$ Statement I is also correct

25. The function $f(x) = \frac{x}{x^2 - 6x - 16}$, $x \in R - \{-2, 8\}$: **[JEE(Main)Jan.2024]**

(A) decreases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(B) increases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(C) decreases in $(-2,8)$ and increases in $(-\infty, -2) \cup (8, \infty)$

(D) decreases in $(-\infty,-2)$ and increases in $(8, \infty)$

फलन $f(x) = \frac{x}{x^2 - 6x - 16}$, $x \in R - \{-2, 8\}$: **[JEE(Main)Jan.2024]**

(A) $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$ में ह्रासमान है

(B) $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$ में वर्धमान है

(C) $(-2,8)$ में ह्रासमान है तथा $(-\infty, -2) \cup (8, \infty)$ वर्धमान है

(D) $(-\infty,-2)$ में ह्रासमान है तथा $(8, \infty)$ में वर्धमान है

Ans. A

Sol. $f(x) = \frac{x}{x^2 - 6x - 16}$

Now

$$f'(x)=\frac{-\left(x^2+16\right)}{\left(x^2-6x-16\right)^2}$$

$f'(x)<0$

Thus f(x) is decreasing in $(-\infty,-2)\cup(-2,8)\cup(8,\infty)$

26. Let $g:R\to R$ be a non constant twice differentiable function such that $g'\left(\frac{1}{2}\right)=g'\left(\frac{3}{2}\right)$. If a real valued function f is defined as $f(x)=\frac{1}{2}[g(x)+g(2-x)]$, then : **[JEE(Main)Jan.2024]**

(A) f" (x) = 0 for no x in (0, 1) (B) f" (x) = 0 for exactly one x in (0, 1)

(C) $f'\left(\frac{3}{2}\right)+f'\left(\frac{1}{2}\right)=1$ (D) f" (x) = 0 for atleast two x in (0, 2)

माना $g:R\to R$ एक परिवर्तनीय तथा दो बार अवकलनीय फलन है और $g'\left(\frac{1}{2}\right)=g'\left(\frac{3}{2}\right)$ है। यदि एक वास्तविक मान फलन f, $f(x)=\frac{1}{2}[g(x)+g(2-x)]$ द्वारा परिभाषित है, तो : **[JEE(Main)Jan.2024]**

(A) (0,1) में कोई भी x नहीं है, जिसके लिए f" (x) = 0 है

(B) (0,1) में ठीक एक x के लिए f" (x) = 0 है

(C) $f'\left(\frac{3}{2}\right)+f'\left(\frac{1}{2}\right)=1$

(D) (0,2) में कम से कम दो x के लिए f" (x) = 0 है

Ans. D

Sol. $f'(x)=\frac{1}{2}\left(g'(x)-g'(2-x)\right)$

Since $f'\left(\frac{1}{2}\right)=f'(1)=f'\left(\frac{3}{2}\right)=0$ (Rolle Theorem)

$\Rightarrow$ f"(x) = 0 for atleast 2 values of $x\in(0,2)$

27. If the function $f:(-\infty,-1]\to(a,b]$ defined by $f(x)=e^{x^3-3x+1}$ is one – one and onto, then the distance of the point P (2b + 4, a + 2) from the line $x+e^{-3}y=4$ is : **[JEE(Main)Jan.2024]**

यदि $f(x)=e^{x^3-3x+1}$ द्वारा परिभाषित फलन $f:(-\infty,-1]\to(a,b]$ एकैकी तथा आच्छादी है, तो रेखा $x+e^{-3}y=4$ से बिन्दु P(2b + 4, a + 2) की दूरी है : **[JEE(Main)Jan.2024]**

(A) $\sqrt{1+e^6}$ (B) $2\sqrt{1+e^6}$ (C) $4\sqrt{1+e^6}$ (D) $3\sqrt{1+e^6}$

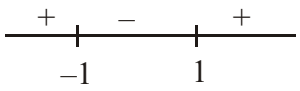
Ans. B

Sol. $f:(-\infty,-1)\to(a,b]$

$f(x)=e^{x^3-3x+1}$

$f'(x)=e^{x^3-3x+1}\left(3x^2-3\right)$

$f'(x)=e^{x^3-3x+1}3(x-1)(x+1)$



$(-\infty,-1]\to\uparrow F^n$

So $x\to-\infty$

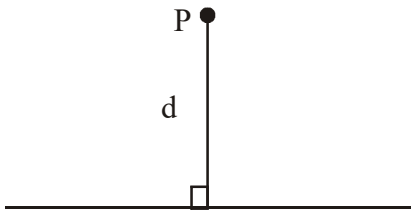
$\Rightarrow f(x)=e^{-\infty}\to 0=a$

$x\to-1$

$\Rightarrow F(-1)=e^{-1+3+1}=e^3=b$

$(a,b]\Rightarrow\left(0,e^3\right]$

$P\left(2e^3+4,2\right)$



$$d=\left|\frac{2e^3+4+e^{-3}(2)-4}{\sqrt{1+\left(e^{-3}\right)^2}}\right|$$

$$d=\frac{2\left(e^3+e^{-3}\right)}{\sqrt{\frac{e^6+1}{e^6}}}$$

$$v=\frac{\frac{2}{e^3}\left(1+e^6\right)}{\frac{\sqrt{1+e^6}}{e^3}}$$

$d=2\sqrt{1+e^6}$

28. If $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2$, $\forall\, x\neq 0$ and $y=9x^2\,f(x)$, then y is strictly increasing in :

[JEE(Main)Jan.2024]

यदि $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2$, $\forall\ x\neq 0$ है तथा $y=9x^2\ f(x)$ है, तो किस अन्तराल में y निरंतर वर्धमान है :

[JEE(Main)Jan.2024]

(A) $\left(0,\frac{1}{\sqrt{5}}\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$ (B) $\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(0,\frac{1}{\sqrt{5}}\right)$

(C) $\left(-\infty,\frac{1}{\sqrt{5}}\right)\cup\left(0,\frac{1}{\sqrt{5}}\right)$ (D) $\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$

Ans. D

Sol. $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2$ ...(1)

Substitute $x\to\frac{1}{x}$

$5f\left(\frac{1}{x}\right)+4f(x)=\frac{1}{x^2}-2$ ...(2)

$(1)\times 5-(2)\times 4$

$\Rightarrow 9f(x)=5x^2-10-\frac{4}{x^2}+8$

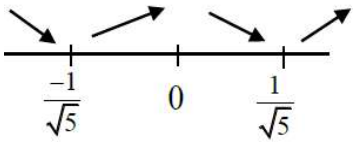
$\Rightarrow 9f(x)=\frac{5x^2-2x^2-4}{x^2}$

$\Rightarrow 9x^2f(x)=5x^4-2x^2-4$

$y=5x^4-2x^2-4$

$\frac{dy}{dx}=20x^3-4x>0$ (given)

$\Rightarrow 4x\left(5x^2-1\right)>0$



$\Rightarrow$ y is increasing in $\left(\frac{-1}{\sqrt{5}},0\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$

29. Let $f(x)=2^x-x^2$, $x\in R$. If m and n are respectively the number of points at which the curves $y=f(x)$ and $y=f'(x)$ intersects the x-axis, then the value of $m+n$ is _____. **[JEE(Main)Jan.2024]**

माना $f(x)=2^x-x^2$, $x\in R$ है। यदि वक्र $y=f(x)$, $y=f'(x)$, x-अक्ष को क्रमशः m तथा n बिन्दुओं पर काटते हैं, तो $m+n$ का मान

है _____. **[JEE(Main)Jan.2024]**

Ans. 5

Sol. $f(x) = 2^x - x^2$

$f(x) = 0 \quad \Rightarrow 2^x = x^2$

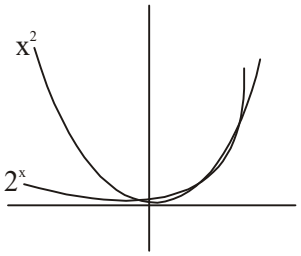
$m = 3$

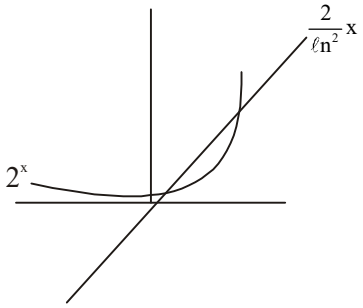
$f'(x) = 2^x \ell n2 - 2x = 0$

$2^x = \frac{2}{\ell n2} x$

$n = 2$

$m + n = 5$





30. In an A.P., the sixth term $a_6 = 2$. If the product $a_1a_4a_5$ is the greatest, then the common difference of the A.P. is equal to : **[JEE(Main)Jan.2024]**

एक A.P. में छठा पद $a_6 = 2$ है। यदि गुणनफल $a_1a_4a_5$ अधिकतम है, तो A.P. का सार्व अंतर बराबर है : **[JEE(Main)Jan.2024]**

(A) 5/8 (B) 8/5 (C) 2/3 (D) 3/2

Ans. B

Sol. Let first term = a

Common difference = d

Given, $a + 5d = 2$ _____(1)

Product (P) = $a_1a_5a_6 = a(a + 4d)(a + 3d)$

From (1)

$P = (2 - 5d)(2 - d)(2 - 2d)$

$\frac{dP}{dd} = (2-5d)(2-d)(-2) + (2-5d)(2-2d)(-1) + (-5)(2-d)(2-2d)$

$= -2[(d - 2)(5d - 2) + (d - 1)(5d - 2) + 5(d - 1)(d - 2)]$

$= -2[5d^2 + 4 - 12d + 5d^2 + 2 - 7d + 5d^2 + 10 - 15d]$

$= -2[15d^2 - 34d + 16]$

$\Rightarrow d = \frac{8}{5} \text{ or } \frac{2}{3}$

at $\left(\frac{8}{5}\right)$, at Product attains maxima

Answer is $\frac{8}{5}$

31. The function $f(x)=2x+3(x)^{\frac{2}{3}}, x \in R$, has : **[JEE(Main)Jan.2024]**

(A) exactly one point of local minima and no point of local maxima

(B) exactly two points of local maxima and exactky one point of local minima

(C) exactly one point of local maxima and exactly one point of local minima

(D) exactly one point of local maxima and no point of local minima

मफलन $f(x)=2x+3(x)^{\frac{2}{3}}, x \in R$: **[JEE(Main)Jan.2024]**

(A) का स्थानीय निम्निष्ठ का मात्र एक बिन्दु है तथा स्थानीय उच्चिष्ठ का कोई बिन्दु नहीं है

(B) के स्थानीय उच्चिष्ठ के मात्र दो बिन्दु है तथा स्थानीय निम्निष्ठ का मात्र एक बिन्दु है

(C) के स्थानीय उच्चिष्ठ का मात्र एक बिन्दु है तथा स्थानीय निम्निष्ठ का मात्र एक बिन्दु है

(D) के स्थानीय उच्चिष्ठ का मात्र एक बिन्दु है तथा स्थानीय निम्निष्ठ का मात्र एक बिन्दु है

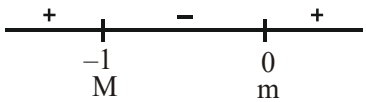
Ans. C

Sol. $f(x)=2x+3(x)^{\frac{2}{3}}$

$f'(x)=2+2x^{\frac{-1}{3}}$

$=2\left(1+\frac{1}{x^{\frac{1}{3}}}\right)$

$=2\left(\frac{x^{\frac{1}{3}}+1}{x^{\frac{1}{3}}}\right)$



So, maxima (M) at $x=-1$ and minima(m) at $x=0$

32. The maximum area of a triangle whose one vertex is at (0, 0) and the other two vertices lie on the curve $y=-2x^2+54$ at points (x, y) and (– x, y), where $y>0$, is : **[JEE(Main)Jan.2024]**

त्रिभुज, जिसका एक शीर्ष (0, 0) है तथा अन्य दो शीर्ष वक्र $y=-2x^2+54$ के बिंदु (x, y) तथा (– x, y) हैं, जहाँ $y>0$ है, का अधि

kd r e {kskQy gS **[JEE(Main)Jan.2024]**

(A) 122 (B) 88 (C) 92 (D) 108

Ans. D

Sol. (–x, y) (x, y)

(0,0) or (0,0)

(–x, y) (x, y)

$\text{Area} = \frac{1}{2}(2x)\left(54-2x^2\right)$

$A = 54x - 2x^3 \Rightarrow \frac{dA}{dx} = 54 - 6x^2$

$x = \pm 3$

Maximum area $A = (54 - 18)3 = 108$

33. Let $f(x) = (x+3)^2(x-2)^3, x \in [-4,4]$. If M and m are the maximum and minimum values of f, respectively in [–4, 4], then the value of M – m is : **[JEE(Main)Jan.2024]**

माना $f(x) = (x+3)^2(x-2)^3, x \in [-4,4]$ हैं। यदि f के [–4, 4] में अधिकतम तथा न्यूनतम मान क्रमशः M तथा m, हैं, तो

M – m का मान है : **[JEE(Main)Jan.2024]**

(A) 392 (B) 600 (C) 108 (D) 608

Ans. D

Sol. $f'(x) = (x+3)^2 \cdot 3(x-2)^2 + (x-2)^3 2(x+3)$

$= 5(x+3)(x-2)^2(x+1)$

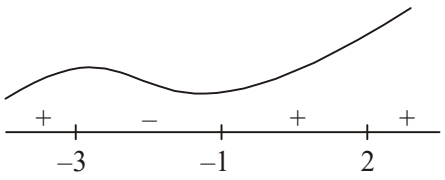
$f'(x) = 0, x = -3, -1, 2$

$f(-4) = -216$

$f(-3) = 0, f(4) = 49 \times 8 = 392$

$M = 392, m = -216$

$M - m = 392 + 216 = 608$



34. Let S = (–1, ∞) and f : S → R be defined as $f(x) = \int_{-1}^{x} \left(e^t - 1\right)^{11} (2t-1)^5 (t-2)^7 (t-3)^{12} (2t-10)^{61} dt$.

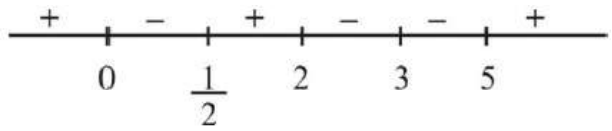
Let p = Sum of squares of the values of x, where f(x) attains local maxima on S, and q = Sum of the values of x, where f(x) attains local minima on S. Then, the value of $p^2 + 2q$ is _____. **[JEE(Main)Jan.2024]**

माना S = (–1, ∞) है तथा $f(x) = \int_{-1}^{x} \left(e^t - 1\right)^{11} (2t-1)^5 (t-2)^7 (t-3)^{12} (2t-10)^{61} dt$ द्वारा परिभाषित फलन

$f : S \to R$ है। माना p = x के उन मानों, जहाँ f, S पर स्थानीय उच्चतम प्राप्त करता है, के वर्गों का योग है तथा q = x के उन मानों, t gkf, S पर स्थानीय न्यूनतम प्राप्त करता है, का योग है। तो $p^2 + 2q$ का मान है_____. **[JEE(Main)Jan.2024]**

Ans. 27

Sol. $f'(x) = (e^x - 1)^{11} (2x - 1)^5 (x - 2)^7 (x - 3)^{12} (2x - 10)^{61}$



Local minima at $x = \frac{1}{2}, x = 5$

Local maxima at $x = 0, x = 2$

$\therefore p = 0 + 4 = 4,\ q = \frac{1}{2} + 5 = \frac{11}{2}$

Then $p^2 + 2q = 16 + 11 = 27$

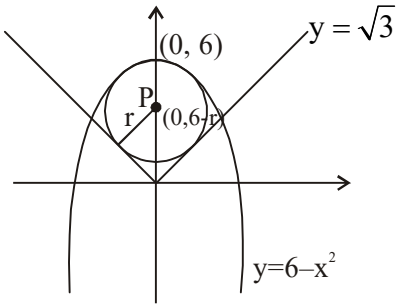
35. Let C be the circle of minimum area touching the parabola $y = 6 - x^2$ and the lines $y = \sqrt{3}|x|$. Then, which one of the following points lies on the circle C ? **[JEE(Main)April,2024]**

माना C न्यूनतम क्षेत्रफल का कोई वृत्त है जो परवलय $y = 6 - x^2$ और रेखा $y = \sqrt{3}|x|$ को स्पर्श कर रहा है। तब निम्नलिखित में से dkSl kfcUnq pUkC पर स्थित होगा? **[JEE(Main)April,2024]**

(A) (1, 2) (B) (2, 2) (C) (1, 1) (D) (2, 4)

Ans. D

Sol.



Equation of circle $x^2 + (y - 6 + r)^2 = r^2$

distance between P and line = r

$\left|\frac{6 - r}{2}\right| = r$

$r = 2$

Equation is $x^2 + (y - 4)^2 = 4$ Point (2, 4) is lie on circle

36. Let maximum and minimum values of $\left(\sqrt{8x - x^2 - 12} - 4\right)^2 + (x - 7)^2$, $x \in R$ be M and m, respectively. Then $M^2 - m^2$ is equal to _____. **[JEE(Main)April,2024]**

माना $\left(\sqrt{8x - x^2 - 12} - 4\right)^2 + (x - 7)^2$, $x \in R$ के उच्चतम तथा निम्नतम मान क्रमशः M तथा m हैं। तो $M^2 - m^2$ बराबर है

Ans. 1600

Sol. $\left(\sqrt{8x-x^2-12}-4\right)^2+(x-7)^2=(y-4)^2+(x-7)^2$

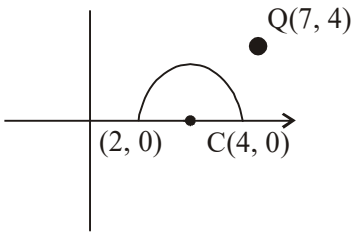
Where $y=\sqrt{8x-x^2-12}>0$

$\Rightarrow x^2+y^2-8x+12=0$

$\Rightarrow (x-4)^2+y^2=4$

Hence, $\left(\sqrt{8x-x^2-12}-4\right)^2+(x-7)^2=(PQ)^2$

Where P (x, y) lies on circle $y=\sqrt{8x-x^2-12}$ and Q (7, 4)



Now, minimum value of $(PQ)^2$ is :

$m=(CQ-2)^2$ where C is centre

$=(5-2)^2=9$

and maximum value of $(PQ)^2$ is :

$M=(7-2)^2+(4-0)^2=25+16=41$

$\Rightarrow \quad M^2-m^2=(41)^2-(9)^2=50\times 32=1600$

37. For the function $f(x)=\sin x+3x-\frac{2}{\pi}\left(x^2+x\right)$, where $x\in\left[0,\frac{\pi}{2}\right]$ consider the following two statements :

(I) f is increasing in $\left(0,\frac{\pi}{2}\right)$

(II) f' is decreasing in $\left(0,\frac{\pi}{2}\right)$

Between the above two statements : **[JEE(Main)April,2024]**

(A) only (I) is true

(B) only (II) is true

(C) both (I) and (II) are true

(D) neither (I) nor (II) is true

फलन

$f(x)=\sin x+3x-\frac{2}{\pi}\left(x^2+x\right)$, जहाँ $x\in\left[0,\frac{\pi}{2}\right]$, के लिए, निम्न दो कथनों पर विचार करें।

(I) f अंतराल $\left(0, \frac{\pi}{2}\right)$ में वर्धमान है

(II) f' अंतराल $\left(0, \frac{\pi}{2}\right)$ में ह्रासमान है

mi jkD nksd Fkuksa ls **[JEE(Main)April,2024]**

(A) केवल (I) सत्य है (B) केवल (II) सत्य है

(C) (I) तथा (II) दोनों सत्य है (D) न तो (I) और न ही (II) सत्य है

Ans. C

Sol. $f(x) = \sin x + 3x - \frac{2}{\pi}(x^2 + x)$, Where $x \in \left[0, \frac{\pi}{2}\right]$

$f'(x) = \cos x + 3 - \frac{2}{\pi}(2x + 1)$

$= \cos x - \frac{4x}{\pi} - \frac{2}{\pi} + 3$

as $x \in \left[0, \frac{\pi}{2}\right]$

$\frac{4x}{\pi} \in [0, 2]$

$\Rightarrow 3 - \frac{2}{\pi} - \frac{4x}{\pi} > 0$

and also $\cos x > 0$ when $x \in \left[0, \frac{\pi}{2}\right]$

$\Rightarrow f'(x) > 0$

$\Rightarrow$ F(x) is increasing

Now, $f''(x) = -\sin x - \frac{4}{\pi} < 0 \; \forall \; x \in \left[0, \frac{\pi}{2}\right]$

Hence, f'(x) is decreasing

$\therefore$ Both statements (I) and (II) are true

38. The interval in which the function $f(x) = x^x, x > 0$, is strictly increasing is : **[JEE(Main)April,2024]**

वह अंतराल, जिसमें फलन $f(x) = x^x, x > 0$, निरंतर वर्धमान है, है : **[JEE(Main)April,2024]**

(A) $\left[\frac{1}{e^2}, 1\right)$ (B) $\left(0, \frac{1}{e}\right]$ (C) $\left[\frac{1}{e}, \infty\right)$ (D) $(0, \infty)$

Ans. C

Sol. $f(x) = x^x$

$f'(x) = x^x (1 + \ln x) \geq 0$

$x > 0 \Rightarrow 1 + \ln x \geq 0$

$x \geq \frac{1}{e}$

$f'(x) \geq 0$ for $x \in \left[\frac{1}{e}, \infty\right)$

39. The number of critical points of the function $f(x) = (x-2)^{\frac{2}{3}}(2x+1)$ is : **[JEE(Main)April,2024]**

फलन $f(x) = (x-2)^{\frac{2}{3}}(2x+1)$ के क्रांतिक बिन्दुओं की संख्या होगी : **[JEE(Main)April,2024]**

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $f(x) = (x-2)^{\frac{2}{3}}(2x+1)$

$f'(x) = 2(x-2)^{\frac{2}{3}} + (2x+1) \times \frac{2}{3}(x-2)^{-\frac{1}{3}}$

$f'(x) = 2(x-2)^{\frac{2}{3}} + \frac{2(2x+1)}{3(x-2)^{\frac{1}{3}}}$

$f'(x) = \frac{6(x-2) + 2(2x+1)}{3(x-2)^{\frac{1}{3}}}$

$f'(x) = \frac{10x-10}{3(x-2)^{\frac{1}{3}}} = \frac{10(x-1)}{3(x-2)^{\frac{1}{3}}}$

Critical points 1,2

$\Rightarrow$ 2 critical points

40. For the function $f(x) = (\cos x) - x + 1$, $x \in R$, between the following two statements :

(S1) $f(x) = 0$ for only one value of x in $[0, \pi]$.

(S2) $f(x)$ is decreasing in $\left[0, \frac{\pi}{2}\right]$ and increasing in $\left[\frac{\pi}{2}, \pi\right]$. **[JEE(Main)April,2024]**

(A) Only (S2) is correct. (B) Only (S1) is correct.

(C) Both (S1) and (S2) are correct. (D) Both (S1) and (S2) are incorrect.

फलन $f(x) = (\cos x) - x + 1, x \in R$ के लिए निचे दिये गये दो कथनों में से :

(S1) $[0, \pi]$ में x के केवल एक मान के लिए $f(x) = 0$ है

(S2) $\left[0, \frac{\pi}{2}\right]$ में f(x) ह्रासमान और $\left[\frac{\pi}{2}, \pi\right]$ में वृद्धमान है **[JEE(Main)April,2024]**

(A) केवल (S2) सत्य होगा (B) केवल (S1) सत्य होगा

(C) (S1) तथा (S2) दोनों सत्य होंगे (D) (S1) तथा (S2) दोनों असत्य होंगे

Ans. B

Sol. $f(x) = \cos x - x + 1$

$f'(x) = -\sin x - 1$

f is decreasing $\forall x \in R$

$f(x) = 0$

$\phi(0) = 2, \phi(\pi) = -\pi$

f is strictly decreasing in $[0, \pi]$ and $f(0).f(\pi) < 0$

$\Rightarrow$ Only one position of $f(x) = 0$

S1 is correct and S2 is incorrect.

41. Let $f : R \to R$ be a thrice diffrentiable function such that $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$. Then, the minimum number of zeroes of $(3f'f'' + ff''')(x)$ is _____. **[JEE(Main)April,2024]**

माना त्रिअवकलनीय एक फलन $f : R \to R$ इस प्रकार है कि $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ है, तथा $f(4) = -2$, तो $(3f'f'' + ff''')(x)$ के न्यूनतम शून्यों की संख्या है _____ **[JEE(Main)April,2024]**

Ans. 5

Sol. $\because f : R \to R$ and $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$ then

f(x) has atleast 4 real roots.

Then f' (x) has atleast 3 real roots and f"(x) has atleast 2 real roots.

$$\frac{d}{dx}\left(f^3 \cdot f''\right) = 3f^2 \cdot f' \cdot f'' + f^3 \cdot f'''$$

$$= f^2\left(3f' \cdot f'' + f \cdot f'''\right)$$

Here $f^3 \cdot f''$ has atleast 6 roots.

Then its differentiation has atleast 5 distinct roots.

42. Let the set of all values of p, for which $f(x)=\left(p^2-6p+8\right)\left(\sin^2 2x-\cos^2 2x\right)+2(2-p)x+7$ does not have any critical point, be the interval (a, b). Then 16ab is equal to _____. **[JEE(Main)April,2024]**

माना p के सभी मानों, जिनके लिए $f(x)=\left(p^2-6p+8\right)\left(\sin^2 2x-\cos^2 2x\right)+2(2-p)x+7$ का एक भी क्रांतिक बिन्दु नहीं g\$d kl eHk;]vUrjky (a, b) है। तो 16ab बराबर है _____. **[JEE(Main)April,2024]**

Ans. 252

Sol. $f(x) = -(p^2-6p+8)\cos 4n + 2(2-p)n + 7$

$f^1(x) = +4(p^2-6p+8)\sin 4x + (4-2p) \neq 0$

$$\sin 4x \neq \frac{2p-4}{4(p-4)(p-2)}$$

$$\sin 4x \neq \frac{2(p-2)}{4(p-4)(p-2)}$$

$p \neq 2$

$$\sin 4x \neq \frac{1}{2(p-4)}$$

$$\Rightarrow \left|\frac{1}{2(p-4)}\right| > 1$$

on solving we get

$$\therefore p \in \left(\frac{7}{2}, \frac{9}{2}\right)$$

Hence $a=\frac{7}{2}, b=\frac{9}{2}$

$\therefore$ 16ab = 252

43. Let $f(x)=3\sqrt{x-2}+\sqrt{4-x}$ be a real valued function. If α and β are respectively the minimum and the maximum values of f, then $\alpha^2+2\beta^2$ is equal to : **[JEE(Main)April,2024]**

माना $f(x)=3\sqrt{x-2}+\sqrt{4-x}$ वास्तविक मूल्य फलन है। यदि α तथा β क्रमशः f के न्यूनतम तथा उच्चतम मान हैं, तो $\alpha^2+2\beta^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 24 (B) 38 (C) 44 (D) 42

Ans. D

Sol. $f'(x)=\frac{3}{2\sqrt{x-2}}-\frac{1}{2\sqrt{4-x}}: f'(x)=0 \qquad \Rightarrow x=\frac{19}{5}\in[2,4]$

$f(2)=\sqrt{2}, f\left(\frac{19}{5}\right)=2\sqrt{5}, f(4)=3\sqrt{2}$

$\therefore \alpha=\sqrt{2}, \beta=2\sqrt{5}$

$\alpha^2 + 2\beta^2 = 2 + 40 = 42$

44. Let a rectangle ABCD of sides 2 and 4 be inscribed in another PQRS such that the vertices of the rectangle ABCD lie on the sides of the rectangle PQRS. Let a and b be the sides of the rectangle PQRS when its area is maximum. Then $(a + b)^2$ is equal to : **[JEE(Main)April,2024]**

माना भुजाओं 2 तथा 4 का एक आयत ABCD एक अन्य आयत PQRS के अंतर्गत इस प्रकार है कि आयत ABCD के शीर्ष आयत PQRS की भुजाओं पर हैं। माना आयत PQRS की भुजाएँ a तथा b है, जब इसका क्षेत्रफल है। तो $(a + b)^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 80 (B) 72 (C) 60 (D) 64

Ans. B

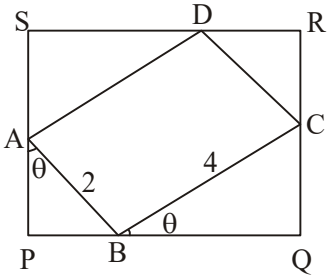
Sol. Area $= (4\cos\theta + 2\sin\theta)(2\cos\theta + 4\sin\theta)$

$= 8\cos^2\theta + 16\sin\theta\cos\theta + 4\sin\theta\cos\theta + 8\sin^2\theta$

$= 8 + 20\sin\theta\cos\theta$

$= 8 + 10\sin 2\theta$

Max Area $= 8 + 10 = 18$ $(\sin 2\theta = 1)$ $\theta = 45°$



$(a + b)^2 = (4\cos\theta + 2\sin\theta + 2\cos\theta + 4\sin\theta)^2$

$= (6\cos\theta + 6\sin\theta)^2$

$= 36(\sin\theta + \cos\theta)^2$

$= 36\left(\sqrt{2}\right)^2$

$= 72$

45. If the function $f(x)=\left(\frac{1}{x}\right)^{2x}; x>0$ attains the maximum value at $x=\frac{1}{e}$ then : **[JEE(Main)April,2024]**

यदि फलन $f(x)=\left(\frac{1}{x}\right)^{2x}; x>0,\ x=\frac{1}{e}$ पर उच्चतम मान प्राप्त करता है, तो : **[JEE(Main)April,2024]**

(A) $e^{2\pi} < (2\pi)^e$ (B) $e^{\pi} > \pi^e$ (C) $e^{\pi} < \pi^e$ (D) $(2e)^{\pi} > \pi^{(2e)}$

Ans. B

Sol. $f(x)=\left(\frac{1}{x}\right)^{2x}$

$f\left(\frac{1}{e}\right)>f\left(\frac{1}{\pi}\right)$

$e^{2/e}>\pi^{2/\pi}$

$e^{\pi}>\pi^{e}$

46. Let $f(x)=4\cos^3 x+3\sqrt{3}\cos^2 x-10$. The number of points of local maxima of f in interval $(0, 2\pi)$ is :
[JEE(Main)April,2024]

माना $f(x)=4\cos^3 x+3\sqrt{3}\cos^2 x-10$, अन्तराल $(0, 2\pi)$ में f के स्थानीय उच्चिष्ठ बिन्दुओं की संख्या होगी :
[JEE(Main)April,2024]

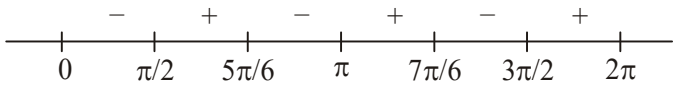
(A) 4 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $f'(x)=-12\cos^2 x\sin x+6\sqrt{3}\cos x\times(-\sin x)$

$=-6\sin x\left[2\cos^2 x+\sqrt{3}\cos x\right]$

$-6\sin x\cos x\left(2\cos x+\sqrt{3}\right)$



local maxima at $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$ so number of local maxima points are 2

47. If the function $f(x)=2x^3-9ax^2+12a^2x+1$, $a>0$ has a local maximum at $x=\alpha$ and a local minimum at $x=\alpha^2$, then α and α^2 are the roots of the equation :
[JEE(Main)April,2024]

यदि फलन $f(x)=2x^3-9ax^2+12a^2x+1$, $a>0$ का $x=\alpha$ पर एक स्थानीय उच्चतम है, α तथा α^2 निम्न में से किस समीकरण d sa/ gS
[JEE(Main)April,2024]

(A) $8x^2+6x-1=0$ (B) $x^2+6x+8=0$ (C) $x^2-6x+8=0$ (D) $8x^2-6x+1=0$

Ans. C

Sol. $f'(x)=6x^2-18ax+12a^2$

$=6(x^2-3a+2a^2)$

$=6(x-a)(x-2a)=0$

$x = a, 2a$

$a = \alpha,$ $\quad 2a = \alpha^2 \quad \Rightarrow \alpha = 0,2$

$a > 0 \quad \therefore \alpha = 2$

$\alpha^2 = 4$

$\therefore x^2 - 6x + 8 = 0$ is the required quadratic equation.

48. A variable line L passes through the point (3, 5) and intersects the positive coordinate axes at the points A and B. The minimum area of the triangle OAB, where O is origin, is : **[JEE(Main)April,2024]**

एक चर रेखा L, बिन्दु (3, 5) से होकर जाती है तथा धनात्मक निर्देशांक अक्षों को बिन्दुओं A और B पर काटती है। यदि O मूलबिन्दु है, r ks=kh OAB का न्यूनतम क्षेत्रफल है : **[JEE(Main)April,2024]**

(A) 35 (B) 25 (C) 40 (D) 30

Ans. D

Sol. As a line passes through P(a, b) cuts the coordinate axis at A and B then area of ΔOAB will be minimum iff P is mid point of AB

$\Rightarrow$ Equation line AB is $\frac{x}{2a} + \frac{y}{2b} = 1$

Hence equation of required line is $\frac{x}{6} + \frac{y}{10} = 1$

$\Rightarrow 5x + 3y = 30$

and area of Δ is $= \frac{1}{2}(6)(10) = 30$

49. Let A be the region enclosed by the parabola $y^2 = 2x$ and the line $x = 24$. Then the maximm area of the rectangle inscribed in the region A is _____. **[JEE(Main)April,2024]**

माना परवलय $y^2 = 2x$ तथा रेखा $x = 24$ से घिरा क्षेत्र A है। तो क्षेत्र A के अंतगर्त आयत का अधिकतम है _____.

[JEE(Main)April,2024]

Ans. 128

Sol. $y^2 = 2x$

$x = 24$

Area $= (24 - x)2y$

$\left(24 - \frac{y^2}{2}\right)2y$

$A = 48y - y^3$

$\frac{dA}{dy} = 48 - 3y^2$, $y = 4$, $x = 8$

Area = $(24 - 8)(2 \times 4)$

$= 128$

50. Let the set of all positive values of λ, for which the point of local minimum of the function $(1 + x(\lambda^2 - x^2))$ satisfies $\frac{x^2 + x + 2}{x^2 + 5x + 6} < 0$, be (α, β). Then $\alpha^2 + \beta^2$ is equal to _____. **[JEE(Main)April,2024]**

माना λ के सभी धनात्मक मानों, जिनके लिए फलन $\frac{x^2 + x + 2}{x^2 + 5x + 6} < 0$ को संतुष्ट करता है, का समुच्चय (α, β) है। तो $\alpha^2 + \beta^2$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 39

Sol. $f(x) = 1 + x(\lambda^2 - x^2)$

$f(x) = -x^3 + \lambda^2 x + 1$

$f'(x) = -3x^2 + \lambda^2$

$x = \pm\frac{\lambda}{\sqrt{3}}$

$-\frac{\lambda}{\sqrt{3}}$ should satisfy the given condition

$\frac{x^2 + x + 2}{x^2 + 5x + 6} < 0$

$\frac{1}{(x+2)(x+3)} < 0$



$x \in (-3, -2)$

$-3 < -\frac{\lambda}{\sqrt{3}} < -2$

$-3\sqrt{3} < -\lambda < -2\sqrt{3}$

$2\sqrt{3} < \lambda < 3\sqrt{3}$

$\left.\begin{aligned}\alpha &= 2\sqrt{3}\\ \beta &= 3\sqrt{3}\end{aligned}\right\}$

$\left(2\sqrt{3}\right)^2 + \left(3\sqrt{3}\right)^2$

$12 + 27$

$= 39$